

Fatima Jinnah Women University, Rawalpindi



CC LAB_1

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Submitted By	Seerat Fatima
Reg. No	2024-BSE-20
Subject	Cloud Computing Lab
Course Code	Cloud Computing (SE-665)
Git Hub username	Seeratfatima97
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Section	B
Submitted To	Sir Waqas Saleem

Task 1 : Create and Configure Your GitHub Account

Explanation: Created a personal GitHub profile using a professional username (seeratfatima97) based on the real name for lab identification and project tracking.

Screenshot 1:



Figure 1.1: GitHub Account Creation & Username Setup

Screenshot 2:

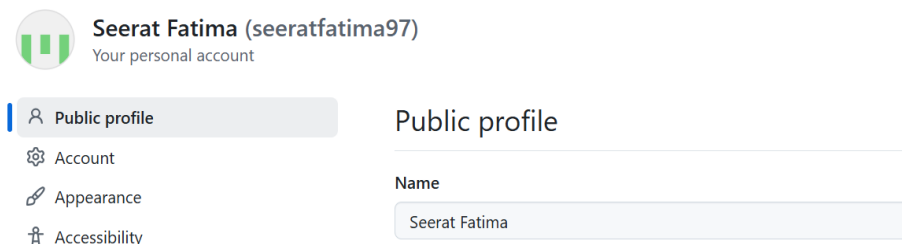


Figure 1.2: Configured Public Profile Settings

Task 2: Register in the Student Marks Portal

Explanation: Authenticated the GitHub account with the Student Marks Portal via Firebase OAuth and submitted the enrollment request for *Cloud Computing (SE-665)*.

Screenshot 1:

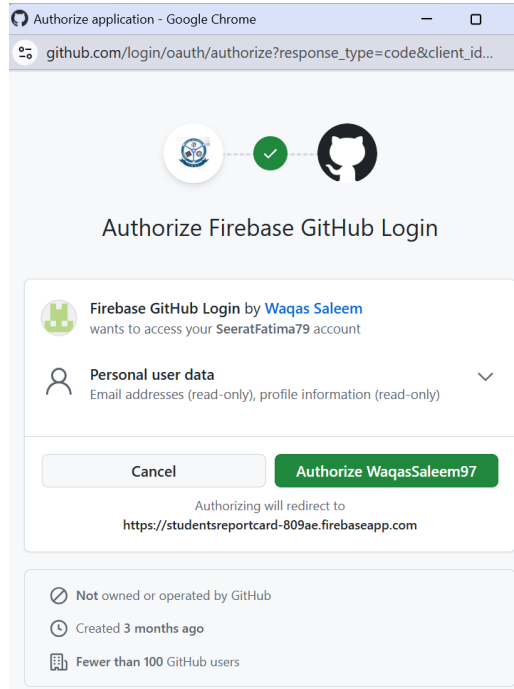


Figure 2.1: Firebase GitHub OAuth Authorization Screen

Screenshot 2:



Figure 2.2: Authenticated GitHub Profile Details inside Portal

Screenshot 3:

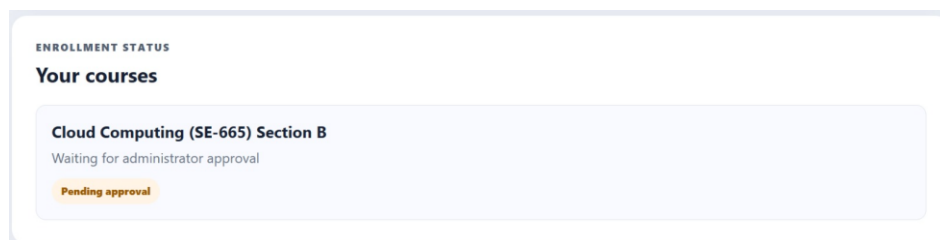


Figure 2.3: Course Enrollment Status Pending Approval

Task 3: Environment Prerequisites Verification

Explanation: Verified all system requirements, including the installation of VMware Workstation Pro, availability of adequate host disk space, and downloading the Ubuntu Server ISO.

Screenshot 1:

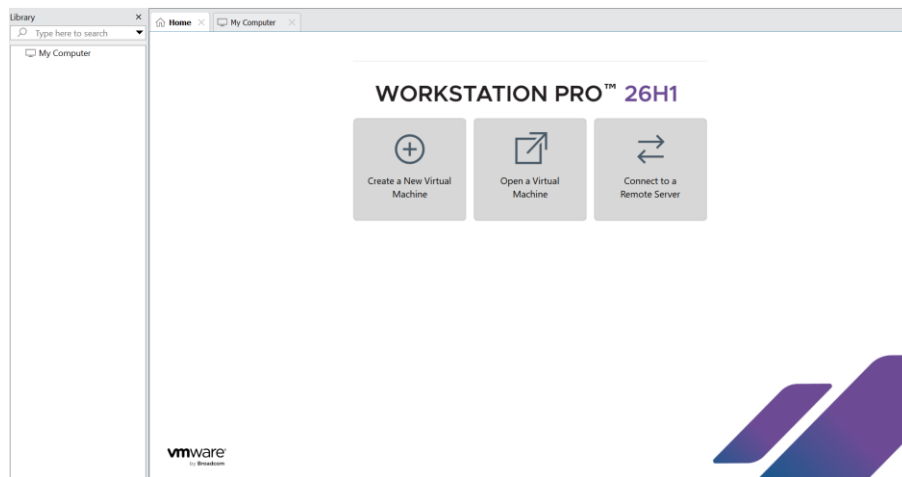


Figure 3.1: VMware Workstation Pro Home Screen

Screenshot 2:

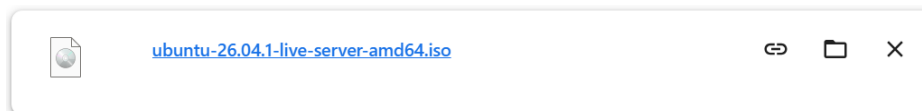


Figure 3.2: Downloaded Ubuntu Server ISO File Confirmation

Screenshot 3:



Figure 3.3: Available System Storage Space on Host Drive

Task 4: Virtual Machine Creation in VMware

Explanation: Initialized a new virtual machine using the Typical (recommended) configuration wizard and mapped the ISO file to boot into the Ubuntu setup.

Screenshot 1:



Figure 4.1: New Virtual Machine Creation Wizard Launch

Screenshot 2:

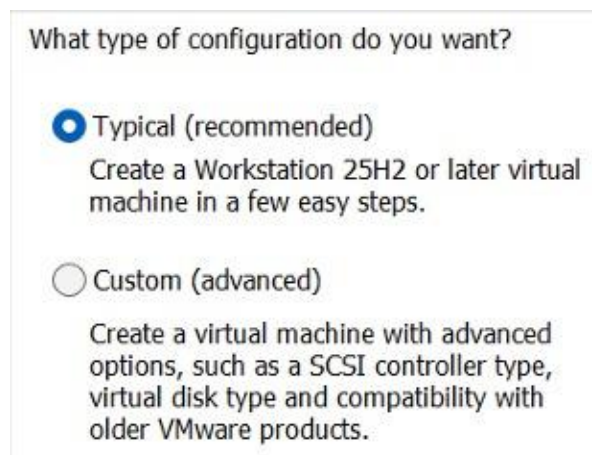


Figure 4.2: Typical Configuration Selection

Screenshot 3:

The screenshot shows the 'New Virtual Machine Wizard' dialog box. The title bar says 'New Virtual Machine Wizard' with a close button. The main heading is 'Name the Virtual Machine' with the instruction 'What name would you like to use for this virtual machine?'. Below this, there is a text field for 'Virtual machine name:' containing 'Ubuntu 64-bit'. Below that is a text field for 'Location:' containing 'C:\Users\User\Documents\Virtual Machines\Ubuntu 64-bit', with a 'Browse...' button to its right. A note below the location field states: 'The default location can be changed at Edit > Preferences.' At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'.

Figure 4.3: Virtual Machine Name and Storage Location Path

Screenshot 4:

The screenshot shows the 'New Virtual Machine Wizard' dialog box at the 'Ready to Create Virtual Machine' step. The title bar says 'New Virtual Machine Wizard' with a close button. The main heading is 'Ready to Create Virtual Machine' with the instruction 'Click Finish to create the virtual machine and start installing Ubuntu 64-bit.' Below this, a section titled 'The virtual machine will be created with the following settings:' contains a list of settings: Name: Ubuntu 64-bit, Location: C:\Users\User\Documents\Virtual Machines\Ubuntu 64-bit, Version: Workstation 25H2 or later, Operating System: Ubuntu 64-bit, Hard Disk: 20 GB, Split, Memory: 2048 MB, Network Adapter: NAT, and Other Devices: 2 CPU cores, CD/DVD, USB Controller, Sound Card. Below the settings list is a 'Customize Hardware...' button. At the bottom, there is a checkbox labeled 'Power on this virtual machine after creation' which is checked. At the very bottom, there are three buttons: '< Back', 'Finish', and 'Cancel'.

Figure 4.4: Final Virtual Machine Hardware Summary

Screenshot 5:

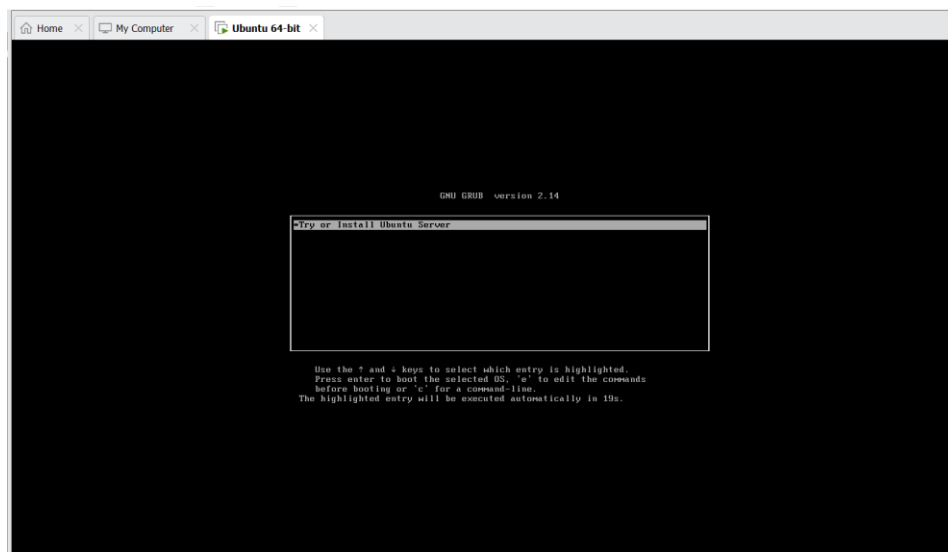


Figure 4.5: GNU GRUB Boot Menu for Ubuntu Server

Task 5: Language Selection

Explanation: Selected English as the system installation language.

Screenshot :

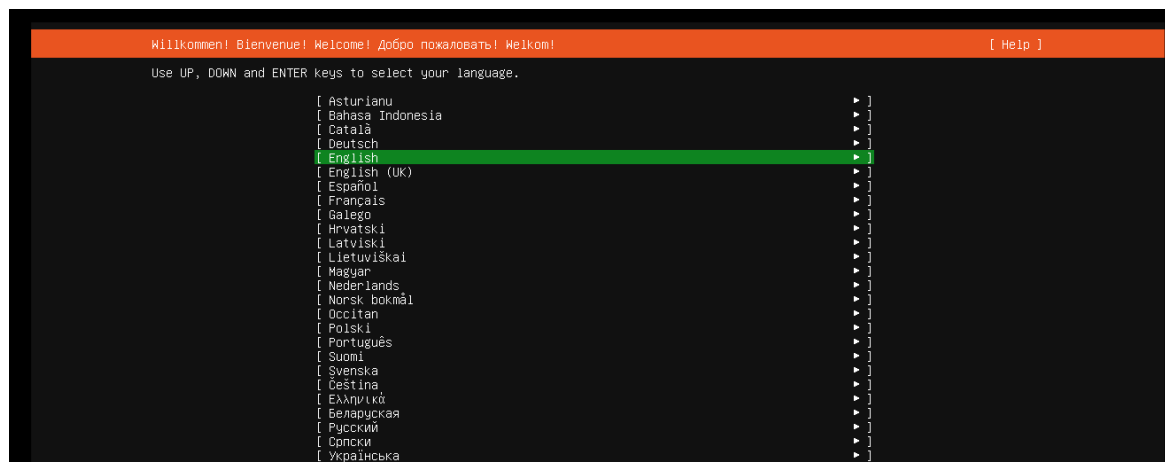


Figure 5.1: Ubuntu Server Installer Language Selection Screen

Task 6: Keyboard Configuration

Explanation: Configured the default English (US) keyboard layout and variant to match host system hardware.

Screenshot 1:

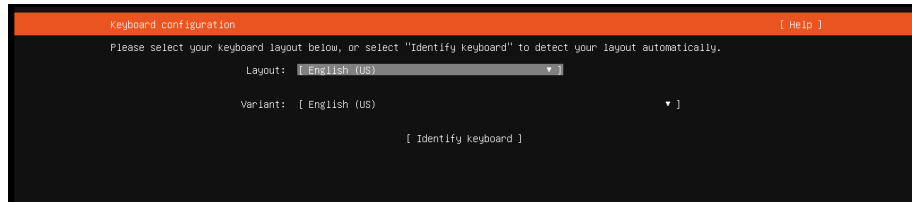


Figure 6.1: Selected English (US) Keyboard Layout

Screenshot 2:

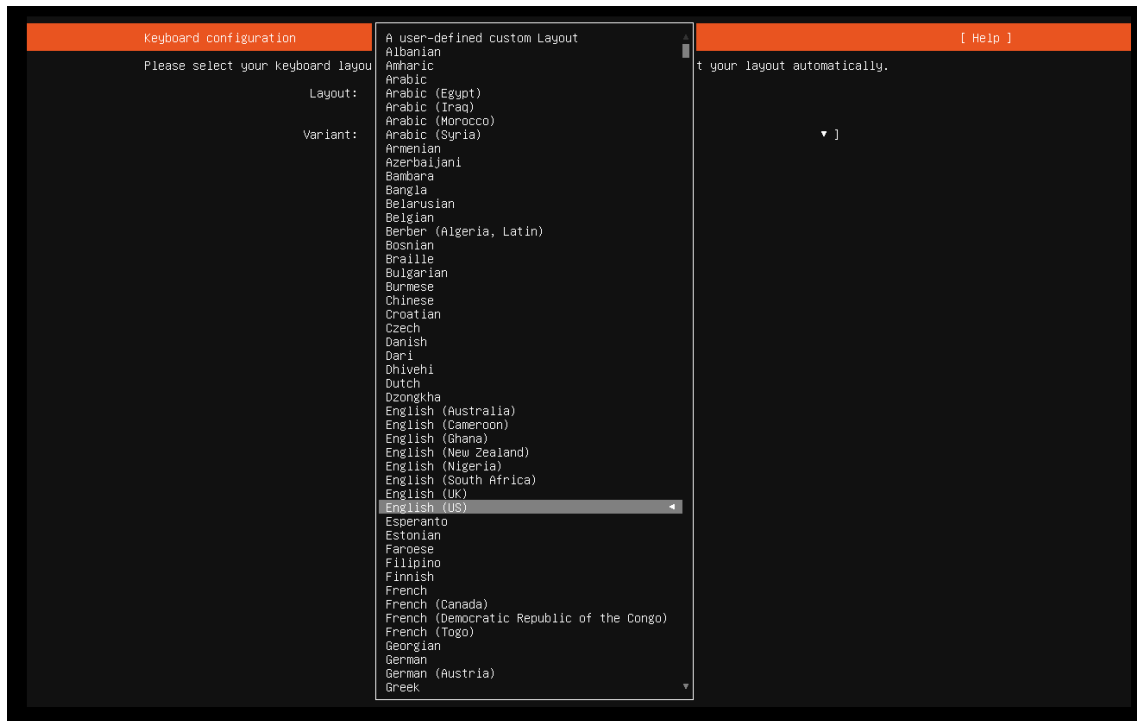


Figure 6.2: Selected Keyboard Variant Dropdown Selection

Task 7: Installation Type Selection

Explanation: Chose the base Ubuntu Server installation option to provide standard server utilities

Screenshot 1:

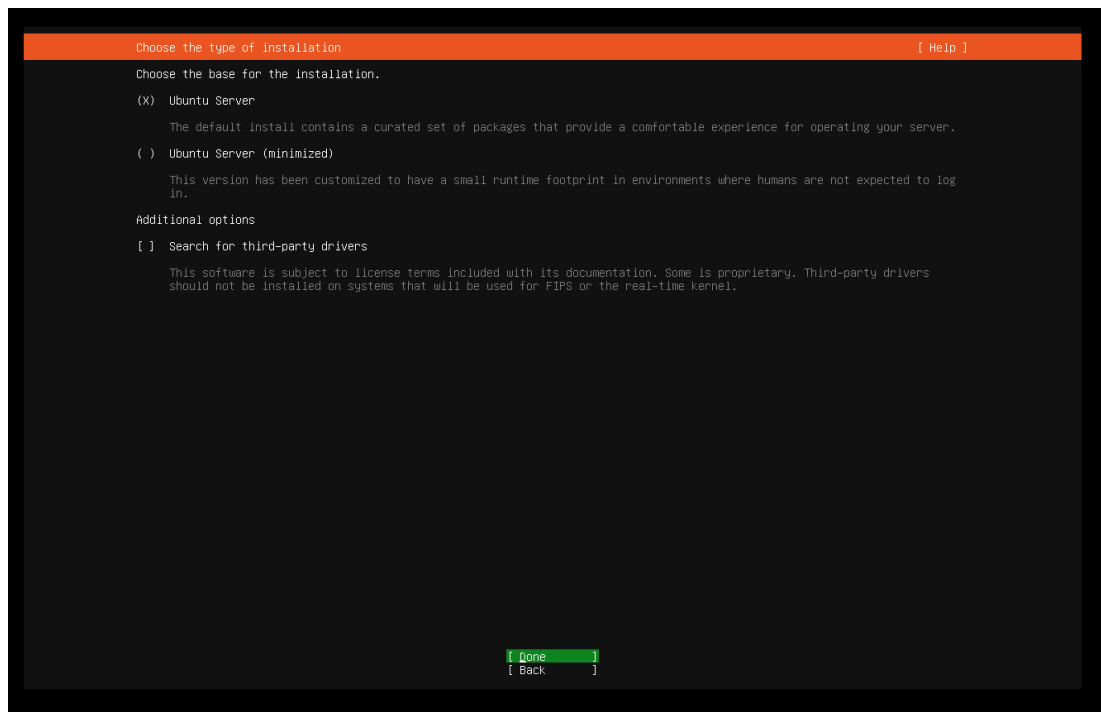


Figure 7.1: Base Ubuntu Server Installation Option Selected

Task 8: Network Configuration

Explanation: Verified network interface detection (ens33) and confirmed dynamic IP assignment via DHCP.

Screenshot 1:

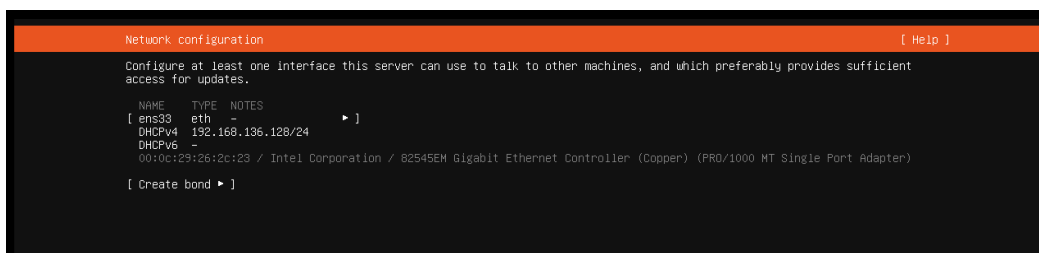


Figure 8.1: Detected Ethernet Interface (ens33)

Screenshot 2:

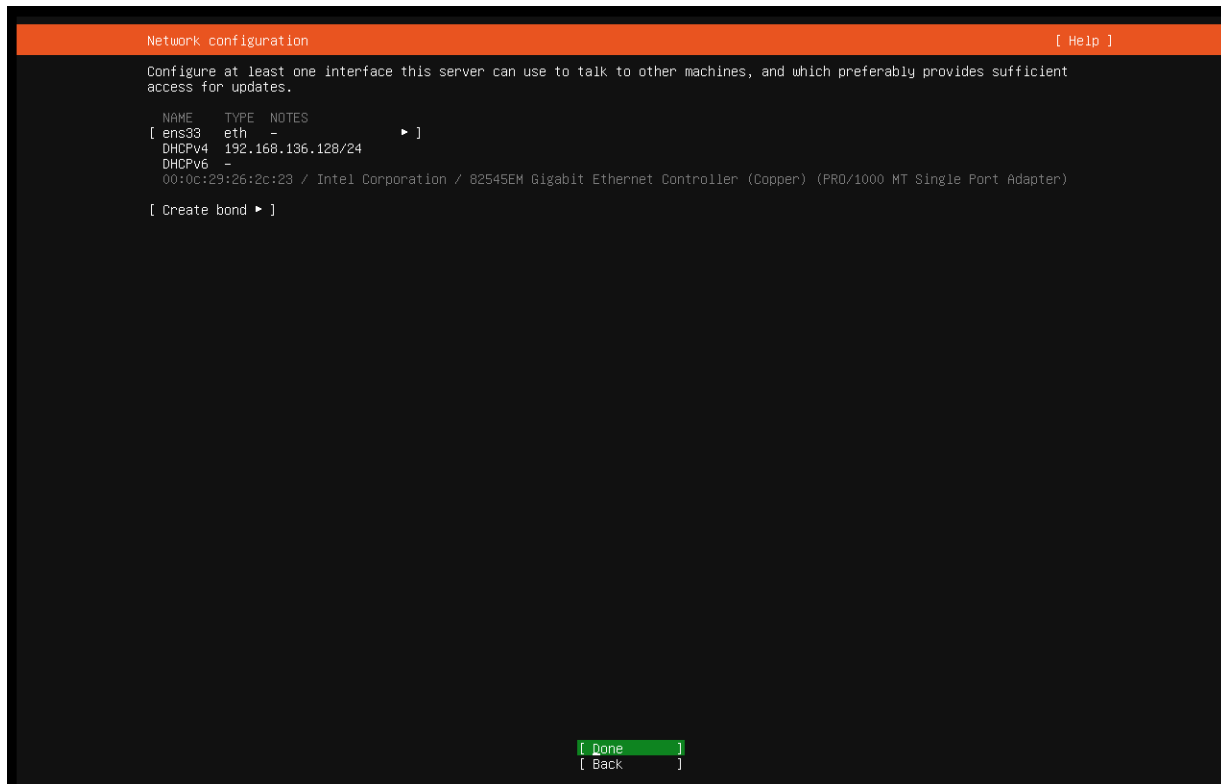


Figure 8.2: Assigned IPv4 Address Information (192.168.136.128/24)

Task 9: Guided Storage Layout Setup

Explanation: Configured guided storage layout to use the entire virtual disk with an LVM (Logical Volume Manager) group.

Screenshot 1:

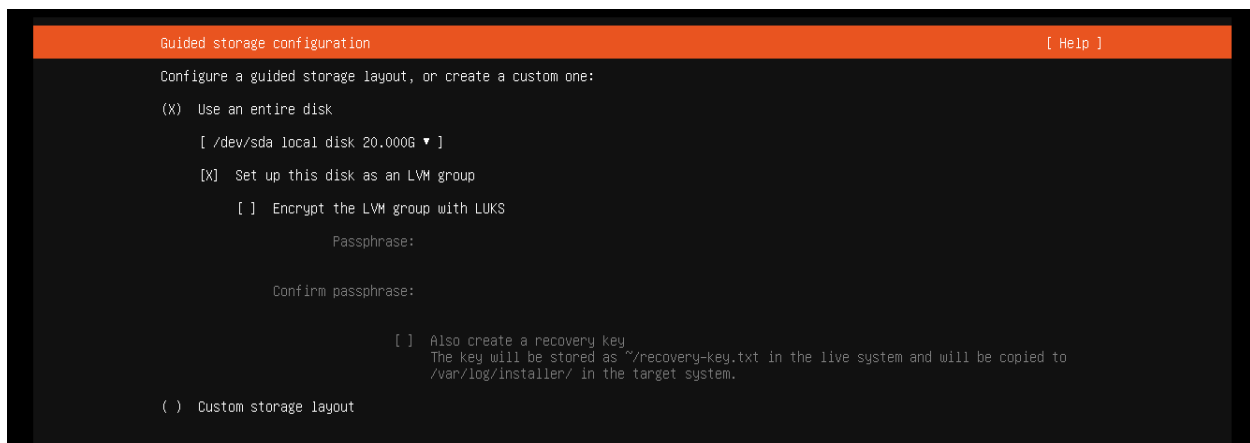


Figure 9.1: Guided Storage Configuration Screen

Screenshot 2:

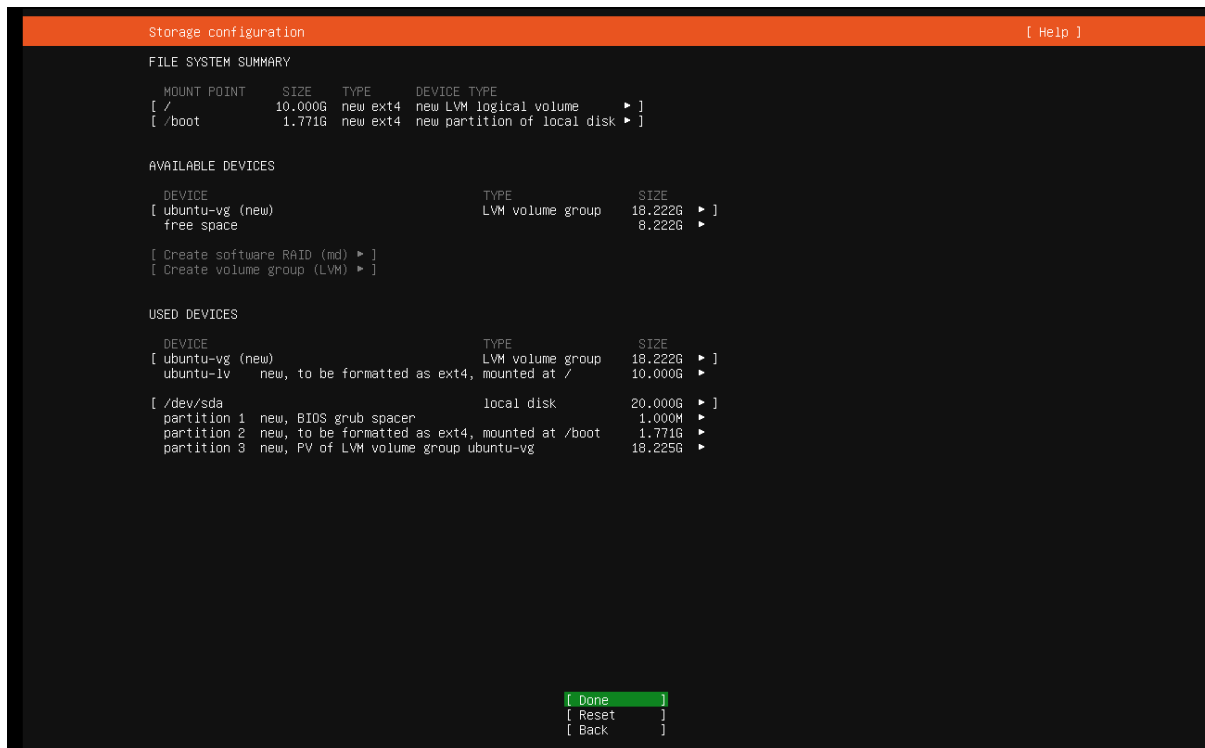


Figure 9.2: Resulting File System Summary and Volume Layout

Task 10: Disk Selection and Hardware Mapping

Explanation: Mapped installation storage to the allocated virtual hard disk (/dev/sda).

Screenshot 1:

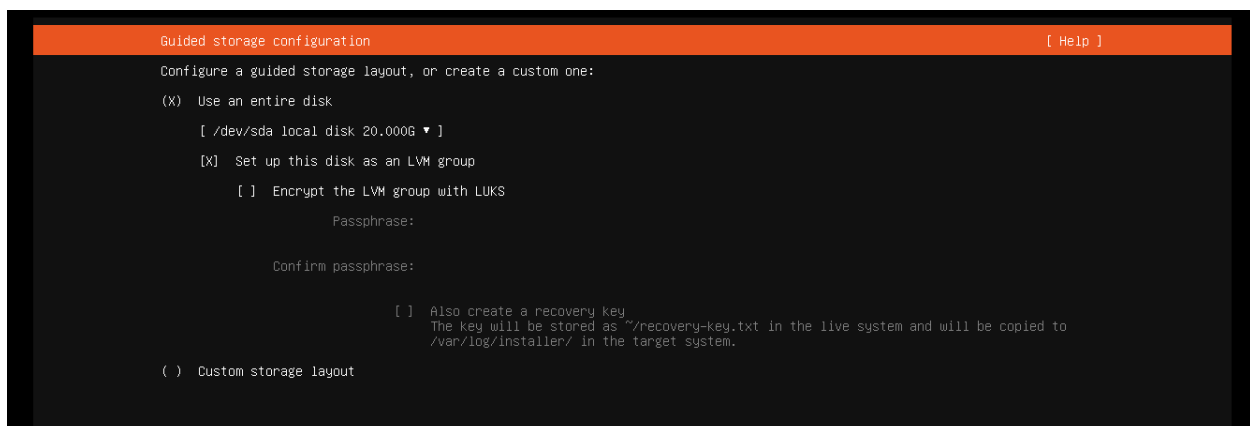


Figure 10.1: Target Local Disk (/dev/sda 20.000G) Selection

Screenshot 2:

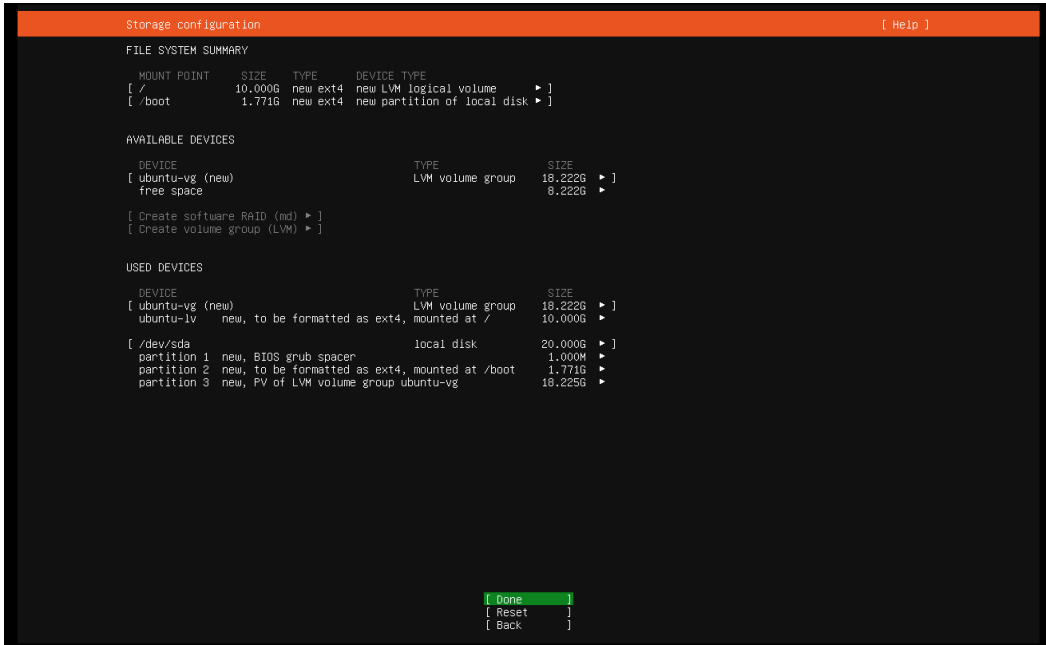


Figure 10.2: Disk Allocation and File System Mount Summary

Task 11: Partition Scheme Verification

Explanation: Reviewed proposed partition layouts, including /boot and LVM logical volume mappings before write-to-disk execution.

Screenshot 1:

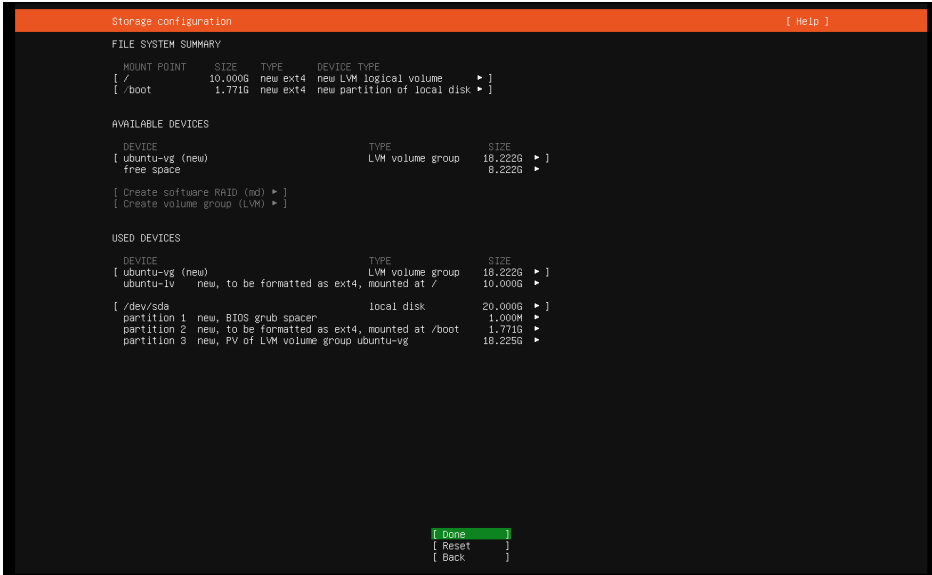


Figure 11.1: Proposed Storage Partition Table Summary

Task 12: Confirm Destructive Storage Action

Explanation: Confirmed the destructive formatting prompt to allow installer access to format disk partitions.

Screenshot 1:

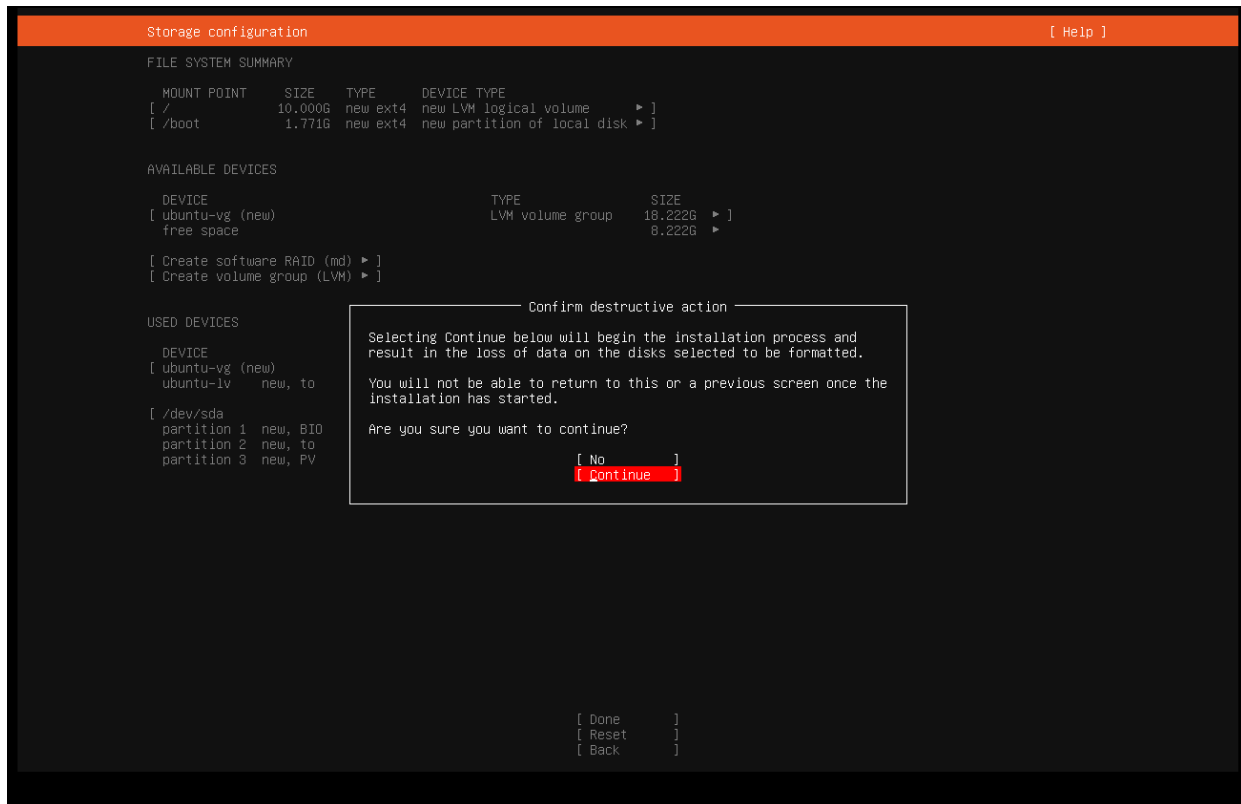


Figure 12.1: Destructive Action Warning Prompt

Screenshot 2:

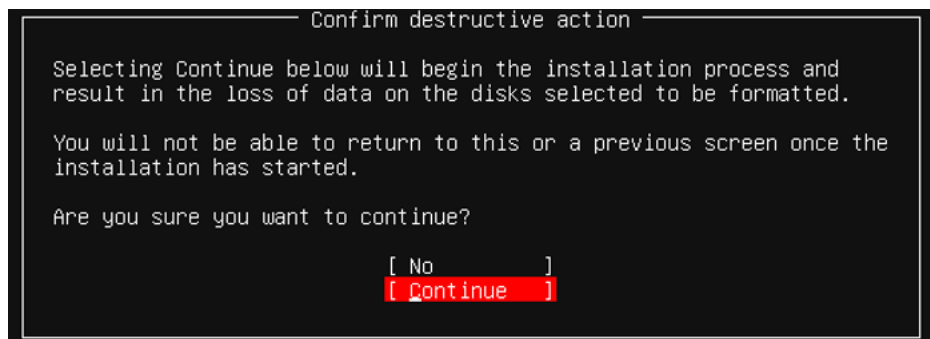
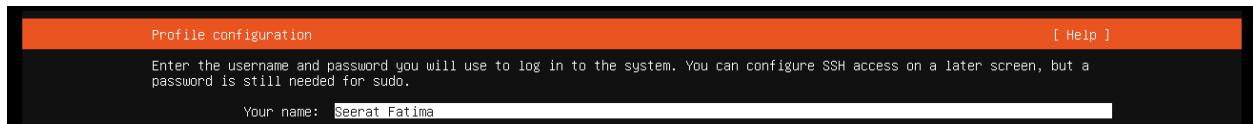


Figure 12.2: Confirmation Choice Selected (Continue)

Task 13: Profile and User Configuration

Explanation: Configured the server hostname (ubuntu) and administrator account credentials (seeratfatima97).

Screenshot 1:



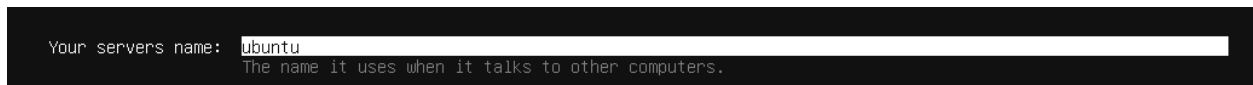
Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Figure 13.1: Entered System User Name

Screenshot 2:



Your servers name:

The name it uses when it talks to other computers.

Figure 13.2: Configured Server Hostname

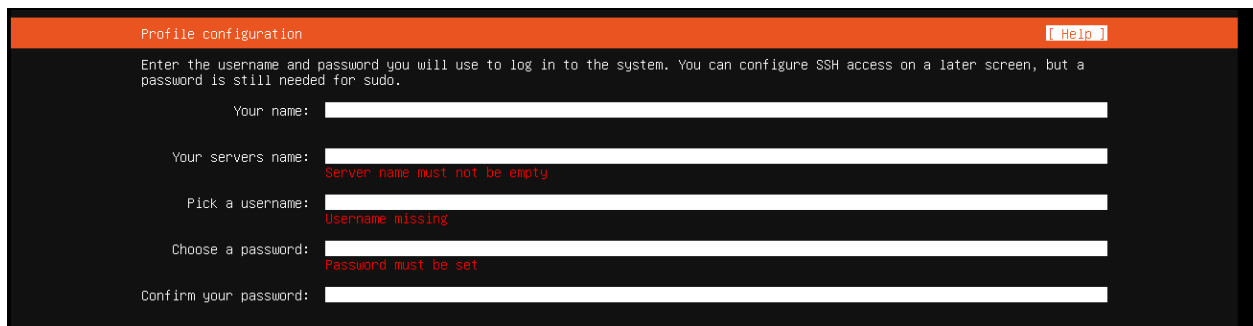
Screenshot 3:



Pick a username:

Figure 13.3: Assigned System Username

Screenshot 4:



Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
Server name must not be empty

Pick a username:
Username missing

Choose a password:
Password must be set

Confirm your password:

Figure 13.4: Profile Configuration Validation Screen

Task 14: System Installation & Package Deployment

Explanation: Monitored real-time system file extraction, core package setup, and OpenSSH server deployment.

Screenshot 1:

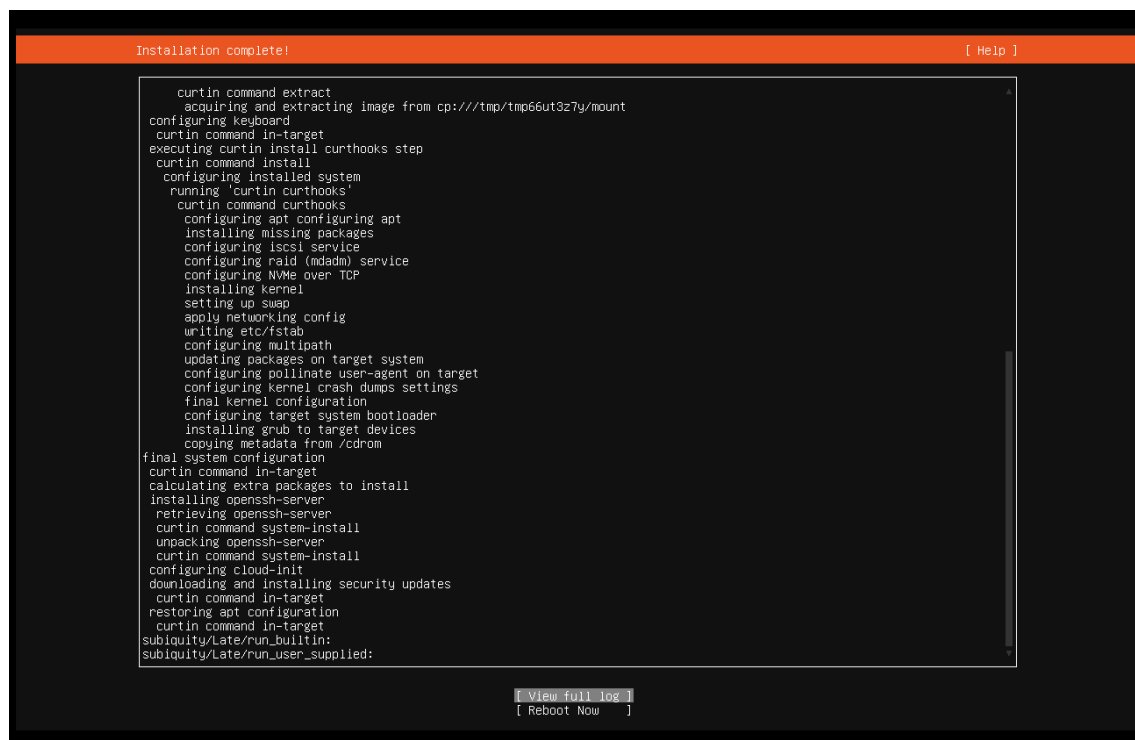


Figure 14.2: System Files Extraction Progress Log and Completion of Server Package Installation

Task 15: Installation Finish & First Boot Login

Explanation: Completed system reboot, reached the TTY login interface, and successfully logged into the local server console.

Screenshot 1:

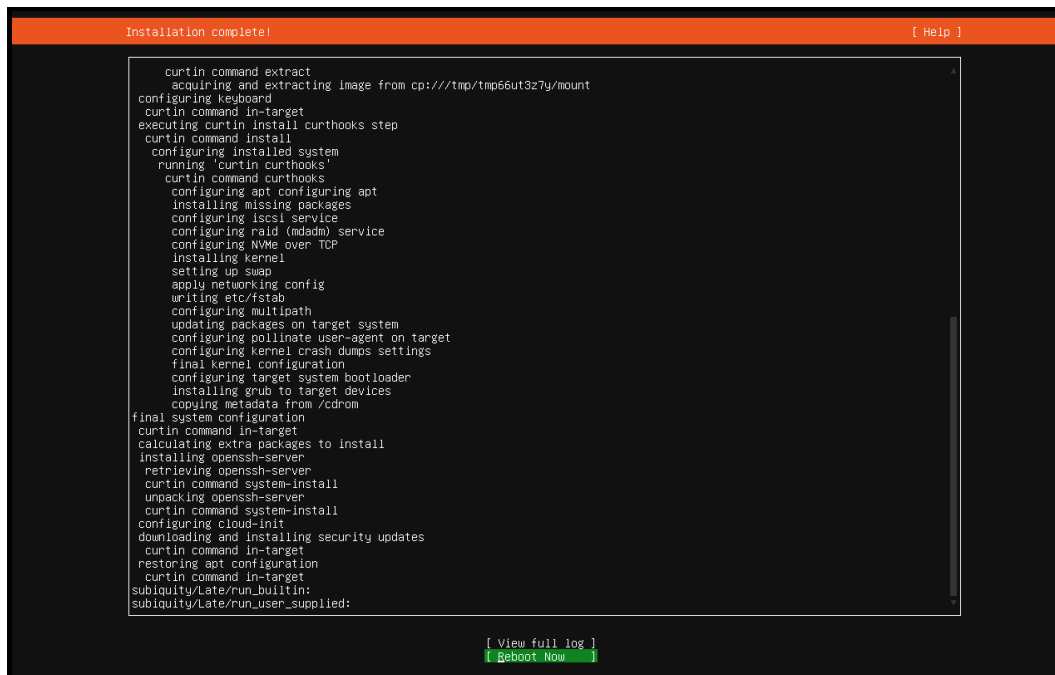


Figure 15.1: Installer Reboot Now Selection

Screenshot 2:

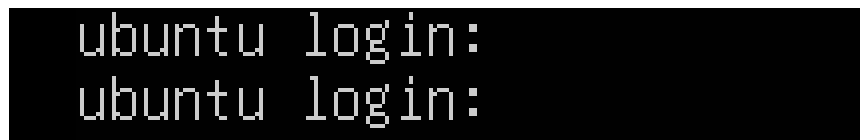


Figure 15.2: Server Console Login Interface Prompt

Screenshot 3:

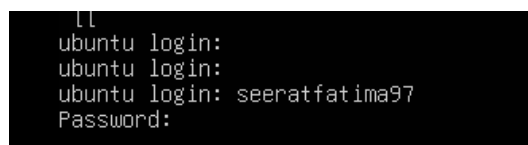


Figure 15.3: Account Credentials Input Screen

Screenshot 4:

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Sat Sep 12 01:45:45 PM UTC 2026

System load:  0.04          Processes:           251
Usage of /:   50.5% of 9.75GB Users logged in:          0
Memory usage: 22%          IPv4 address for ens33: 192.168.136.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

seeratfatima97@ubuntu:~$
```

Figure 15.4: Successful Local Terminal Session and System Banner

Task 16: Verify Server IPv4 Address

Explanation: Executed the `ip addr` command to retrieve active network configuration details and note down the server's private IPv4 address (192.168.136.129).

Screenshot 1:

```
seeratfatima97@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:26:2c:23 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29262c23
    inet 192.168.136.129/24 metric 100 brd 192.168.136.255 scope global dynamic ens33
        valid_lft 1136sec preferred_lft 1136sec
    inet6 fe80::20c:29ff:fe26:2c23/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
seeratfatima97@ubuntu:~$
```

Figure 16.1: Output of `ip addr` Command

Screenshot 2:

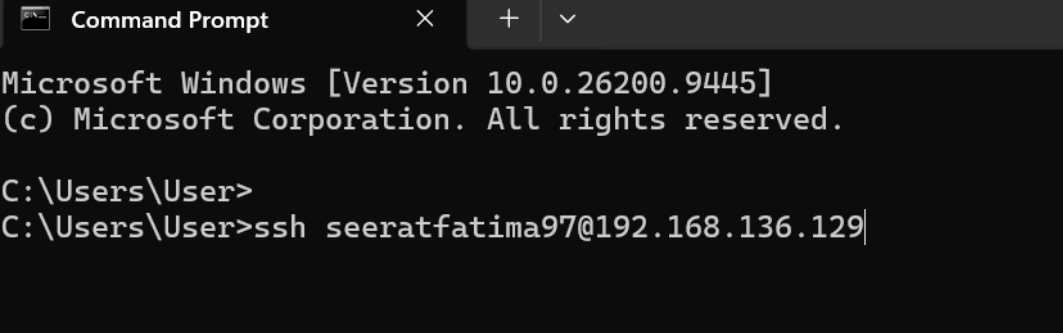
```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000  
    link/ether 00:0c:29:26:2c:23 brd ff:ff:ff:ff:ff:ff  
    altname enp2s1  
    altnam enx000c29262c23  
    inet 192.168.136.129/24 metric 100 brd 192.168.136.255 scope global dynamic ens33  
        valid_lft 1136sec preferred_lft 1136sec  
    inet6 fe80::20c:29ff:fe26:2c23/64 scope link proto kernel_ll  
        valid_lft forever preferred_lft forever  
seeratfatima97@ubuntu:~$
```

Figure 16.2: Highlighted IPv4 Address (192.168.136.129) on ens33 Interface

Task 17: Connect to Ubuntu Server via Windows SSH

Explanation: Established a remote SSH connection to the Ubuntu server from Windows Command Prompt, accepted host authenticity key fingerprints, and verified terminal prompt access.

Screenshot 1:



```
Microsoft Windows [Version 10.0.26200.9445]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\User>  
C:\Users\User>ssh seeratfatima97@192.168.136.129
```

Figure 17.1: Typed SSH Connection Command (ssh seeratfatima97@192.168.136.129)

Screenshot 2:

```
The authenticity of host '192.168.136.129 (192.168.136.129)' can't be established.  
ED25519 key fingerprint is SHA256:sCrWkqds70DoCCmjEas2qQzHbIPoNAlaIVfPrPc46Uw.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.136.129' (ED25519) to the list of known hosts.
```

Figure 17.2: Host ED25519 Fingerprint Key Acceptance

Screenshot 3:

```
C:\Users\User>ssh seeratfatima97@192.168.136.129  
seeratfatima97@192.168.136.129's password:
```

Figure 17.3: Remote User Password Authentication Input

Screenshot 4:

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Sat Sep 12 02:13:04 PM UTC 2026

System load:  0.08           Processes:            219
Usage of /:   50.6% of 9.75GB Users logged in:           0
Memory usage: 21%           IPv4 address for ens33: 192.168.136.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

seeratfatima97@ubuntu:~$ |
```

Figure 17.4: Established SSH Remote Session Showing Active Remote Prompt (seeratfatima97@ubuntu:~\$)

-----THE END-----